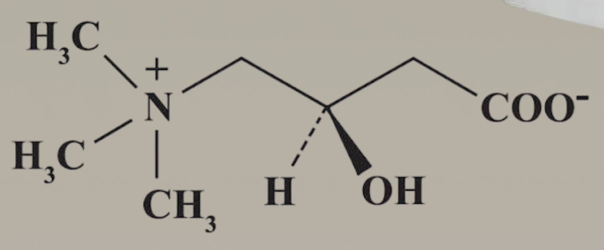
3

L-CARNITINE

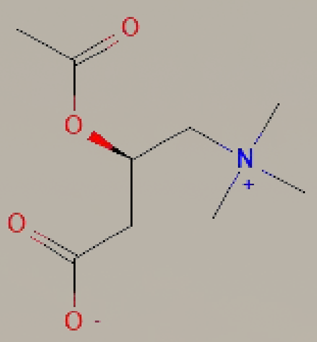
A Sulfur-Containing Amino Acid

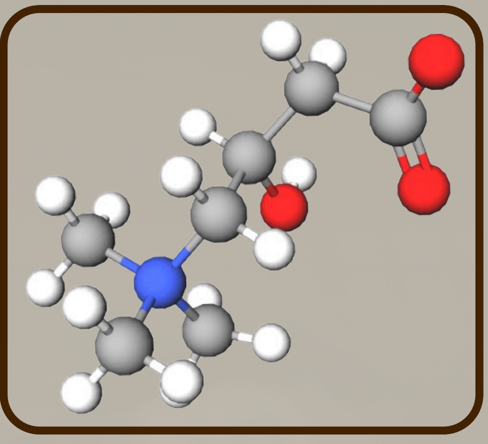
CHEMICAL IDENTITY

- **IUPAC Name:** (3R)-3-hydroxy-4-(trimethylazaniumyl)butanoate
- **Molecular Formula:** $C_7H_{15}NO_3$
- **Molar Mass:** 161.20 g/mol
- **CAS No.:** 541-15-1
- **PubChem CID:** 10917



STRUCTURE (L-Carnitine)





INTERESTING FACT

L-carnitine plays a crucial role in transporting long-chain fatty acids into mitochondria, where they can undergo β-oxidation to produce energy

L-carnitine temporarily forms acylcarnitines with fatty-acyl groups. This allows long-chain fatty acids to cross the inner mitochondrial membrane before being released for β-oxidation.

KEY PROPERTIES & DATA

Appearance: White crystalline solid
Molecular weight: 161.20 g/mol
Formula: $C_7H_{15}NO_3$
Hydrogen-bond donors: 1
Hydrogen-bond acceptors: 3
Topological polar surface area: 60.4 Å²
Chemical character: Water-soluble, zwitterionic compound

USES & VALUE

Biological role

- Plays an essential role in cellular energy metabolism.
- Helps support energy production from fatty acids.
- Used medically to manage certain forms of carnitine deficiency.

Commercial/consumer use

- Commercially produced for use in supplement and pharmaceutical applications.
- Its value comes from its established role in fatty-acid metabolism and its applications in specific nutritional and medical products.

SOURCES

Red meat, Dairy Products (Milk, Butter, Cream), Supplements and Skin Care Products

RISKS AND SAFETY

- *Carnitine naturally occurs in foods and the human body.*
- *Excessive supplemental intake can cause nausea, vomiting, diarrhea, and a fishy body odor.*
- *Certain medical conditions may increase the risk of adverse effects.*
- *Also investigated the conversion of carnitine by gut bacteria into trimethylamine N-oxide (TMAO); health implications remain an area of research.*

CHEMICAL ENGINEERING CONNECTION

- Reaction Engineering: Optimizes conditions for L-carnitine production.
- Separation and Purification: Removes impurities and unwanted reaction products.
- Crystallization: Can be used to recover and purify the solid product.
- Filtration & Drying: Separates crystals and removes residual moisture.
- Mass Balances: Tracks reactants, products, and waste throughout the process.
- Process Optimization: Improves yield, purity, and production efficiency.
- Quality Control: Ensures chemical identity, purity, and product consistency.